

# Noninvasive genetic analyses for estimating population size and genetic diversity of the remaining Far Eastern leopard (*Panthera pardus orientalis*) population

Taro Sugimoto · Vladimir V. Aramilev ·  
Linda L. Kerley · Junco Nagata · Dale G. Miquelle ·  
Dale R. McCullough

Received: 21 July 2013 / Accepted: 16 December 2013 / Published online: 28 December 2013  
© Springer Science+Business Media Dordrecht 2013

**Abstract** Understanding and monitoring the population status of endangered species is vital for developing appropriate management interventions. We used noninvasive genetic analyses to obtain ecological and genetic data on the last remaining Far Eastern leopard population in the world. During seven winters from 2000–2001 to 2007–2008, we collected feces, hair, and saliva from most of the leopard habitat. Of the 239 leopard samples collected during the study period, 155 were successfully genotyped at 13 microsatellite loci and 37 individuals (18 males and 19 females) were identified. Population size estimates based on the Capwire model were 28 (95 % CI 19–38) in 2002–03 and 26 (95 % CI 13–33) in 2007–2008. The leopard population had a low level of genetic diversity (expected and observed

heterozygosity = 0.43; average number of alleles per locus = 2.62), and effective population size was estimated to be low ( $N_e = 7\text{--}16$ ) by two genetic-based methods. We observed little improvement in the genetic diversity during the study period and did find an indication of allele loss compared with individuals from the mid-1990s, suggesting that the remaining population will continue to suffer loss of genetic diversity. Given the small population size and the low genetic diversity, with little expectation of replenishment of the genetic variation by natural immigration, successful expansion of available habitat and development of a second population based on captive individuals may be crucial for persistence of this leopard subspecies in the wild.

**Keywords** Carnivore conservation · Endangered species · Fecal DNA · Low genetic diversity · Noninvasive genetic sampling · *Panthera pardus*

T. Sugimoto (✉)  
Faculty of Environmental Earth Science, Hokkaido University,  
Sapporo, Hokkaido 0600810, Japan  
e-mail: tasugi@ees.hokudai.ac.jp

V. V. Aramilev  
Pacific Institute of Geography, Far Eastern Branch of the  
Russian Academy of Sciences, Vladivostok 690041, Russia

L. L. Kerley  
Zoological Society of London, Regent's Park, London, UK

J. Nagata  
Forestry and Forest Products Research Institute,  
Tsukuba 3058687, Japan

D. G. Miquelle  
Russia Program, Wildlife Conservation Society, Bronx,  
NY 10460, USA

D. R. McCullough  
Department of Environmental Science, Policy, and Management,  
and Museum of Vertebrate Zoology, University of California,  
Berkeley, Berkeley, CA 94720, USA

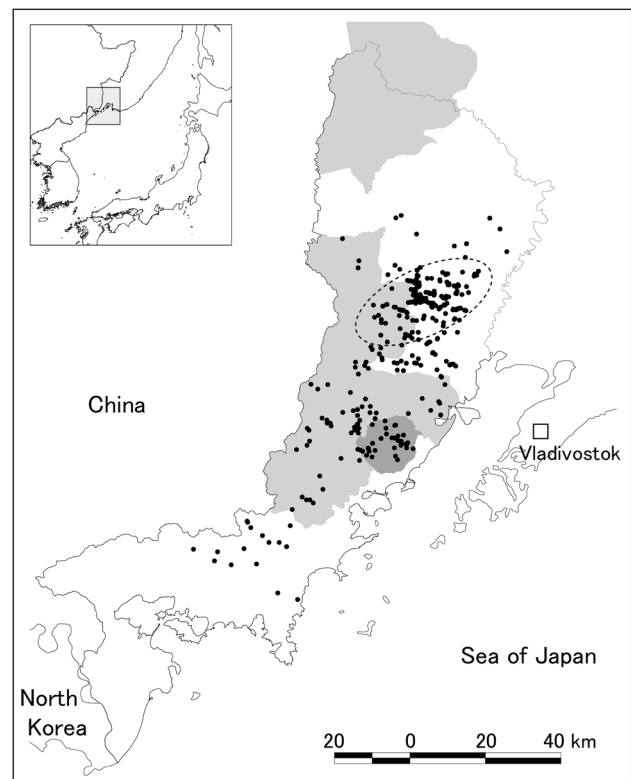
## Introduction

Monitoring temporal changes in the population size and genetic diversity of endangered species is critical for developing conservation strategies and evaluating current efforts. This is particularly important for small and isolated populations because they are vulnerable to the effects of genetic drift and demographic stochasticity. Population monitoring of elusive species based on direct field-based observations is difficult, but noninvasive genetic sampling allows the monitoring of key population parameters without the need to observe or capture animals (Schwartz et al. 2007). However, the error-prone nature of noninvasive genetic analyses means that DNA experiments should be carefully performed (Taberlet et al. 1999), particularly when investigating species with low genetic diversity

because increasing the number of loci to improve the resolution of individual identification will increase the probability of genotyping errors (Waits and Paetkau 2005). The number of studies applying noninvasive genetic sampling to long-term population monitoring has been limited (e.g. De Barba et al. 2010; Koelewijn et al. 2010; Adams et al. 2011), but such “genetic monitoring” is expected to be an ever more valuable tool for the conservation and management of endangered species (Schwartz et al. 2007).

The Far Eastern leopard (*Panthera pardus orientalis*), the northeastern-most subspecies of leopard, is one of the most endangered felid subspecies in the world. Historically, the range of Far Eastern leopards included northeast China, the Korean peninsula, and the southernmost part of the Russian Far East (Pocock 1930; Nowell and Jackson 1996). In the twentieth century, diverse anthropogenic impacts such as logging, poaching, and human developments led to a dramatic decrease in the population size and distribution (Miquelle and Pikunov 2003). Three populations were found when the first robust survey was conducted in Russia in 1974, with an estimated total population size of 38–46 (Abramov and Pikunov 1974). In a 1985 survey, two of these three populations had apparently already disappeared (Pikunov and Korkishko 1985). Thus, the population in Southwest Primorskii Krai (Province) is all that remains with estimates of population size ranging from 22 to 50 (Aramilev and Fomenko 2000; Pikunov et al. 2003) (Fig. 1). In China, historical information on the leopard’s status is rare. In a 1998 survey, several leopard tracks were found along the Russian border in Jilin Province, although these may have represented dispersers from the Russian side (Yang et al. 1998), whereas there was no evidence of the leopards in Heilongjiang Province (Sun et al. 1999). There have been no subsequent reports of the leopard’s presence in China. However, several leopards have recently been photographed using camera traps in Jilin Province (Wildlife Conservation Society 2012; Wildlife Conservation Society Press Release, 27 November 2013), just across the border from the Russian population ranging as far as 30 km from the border, which may suggest recolonization of its historical range.

Surveys of leopards in Russia have traditionally been conducted using a track count along standardized routes, which attempts to estimate population size by distinguishing the numbers of individuals based on their track size, the distance between tracks, and the age of tracks (Miquelle et al. 2006). While this method provides valuable information on the distribution of the population, there exist many errors associated with changes in track size, snow degradation, double-counting of the same individual on different transects, and varying interpretations of track data. Noninvasive genetic sampling is therefore expected to provide a useful independent estimate to compare with the traditional snow-track count. This approach also allows an



**Fig. 1** Map of Southwest Primorskii Krai, Russian Far East with locations of the noninvasive genetic samples collected during seven winters (2000–2001 to 2007–2008). An area surrounded by the dotted line is referred to as the high track density area where tracks of leopards and tigers were frequently found by the snow-track counts (Pikunov et al. 2003). The shaded area indicates protected areas and the darker area indicates the Kedrovaya Pad Nature Reserve

evaluation of genetic diversity. Analyses by Uphyrkina et al. (2002) of seven individuals captured in Southwest Primorskii Krai between 1993 and 1996 indicated the lowest level of genetic diversity among seven leopard subspecies, but a larger sample size over a larger proportion of this subspecies range may provide a more accurate assessment of genetic diversity.

The present study conducted noninvasive genetic analyses to obtain ecological and genetic data on the remaining Far Eastern leopard population. In particular, we aimed to identify individuals, estimate population size, and assess genetic diversity. In addition, we considered their spatial ecology based on the spatial distribution of multiple samples per individual.

## Materials and methods

### Sampling and DNA extraction

We collected noninvasive genetic samples (feces, hair, and saliva) from the leopard habitat in Southwest Primorskii

Krai, Russian Far East (Fig. 1) during seven winters (2000–2001 through 2007–2008, except 2003–2004) (Table 1). This area shares borders with China in the west and North Korea in the south, and is dominated by the eastern slopes of the east Manchurian Mountains. The dominant habitat types were deciduous broad-leaved forest dominated by Mongolian oak (*Quercus mongolica*) and mixed conifer-deciduous forest dominated by Korean pine (*Pinus koraiensis*), larch (*Larix komarovii*), black fir (*Abies holophylla*), and birches (*Betula costata*, *B. lanata*, and others). Three protected areas of varying status were combined and expanded into a single management system known as the “Land of the Leopard National Park” in 2012. Most of the settlements in this regime are located along the coast and outside the protected area. A federal highway runs through Southwest Primorskii Krai with multiple side roads and forest roads (mostly logging roads) latticing the area, and thus, most leopard habitat is easily accessible. The majority of samples were collected between December and March, but a few (8 %) that were obtained outside this period (April, May, October, or November) were allocated to the nearest winter season. In addition to project personnel, samples were also collected by rangers in local nature reserves and wildlife refuges, state hunting inspectors, and local hunters. We did not establish a priori transects for sampling, but instead, we surveyed the trails used by leopards or Amur tigers (*Panthera tigris altaica*) on a regular basis, as determined by snow-track surveys and long-term experience of the field teams, thereby facilitating more efficient collection of samples. Feces were collected opportunistically or by following the tracks of large cats. We employed scat detection dogs (MacKay et al. 2008; Kerley 2010) during two winters (2005–2006 and 2006–2007) and collected 16 % of the samples with their assistance (Table 1). Hair samples were collected from trees, bushes, fences around deer farms, and military barbed wire. During 2004–2005, hair samples were also collected using hair traps (Table 1), where a small synthetic carpet was placed on trees or the ground in

a high track density area (Fig. 1). Details of the hair trap sampling method can be found in Sugimoto et al. (2012). Saliva samples were collected using cotton swabs from wounds of prey killed such as sika deer and roe deer. For all but 5 % of the samples, the locations were recorded with a GPS or on a 1:100,000 topographic map with an estimated error of <1 km.

Sampling intensity and area covered varied among winters because of differences in the number of people involved and the efforts of each individual. During 2001–2002, 2002–2003, 2006–2007, and 2007–2008, approximately 10–30 people were involved with sampling and they covered the majority of known leopard habitat, except the southern edge of leopard distribution where the track density was quite low (Pikunov et al. 2003, 2009; Hebblewhite et al. 2011) (Table 1). The high track density area was not included only for the 2006–2007 sampling period (Table 1). In each of the 4 winters, approximately 20–30 days (8 h per day) were spent collecting samples during several months in winter. During the other winters (2000–2001, 2004–2005, and 2005–2006), project personnel collected a relatively small number of samples from a limited number of sites (Table 1).

Most samples (84 %), including hair and saliva samples, were placed in ziplock bags at ambient temperature and transported to a freezer (−20 °C) at a field base normally within 1–4 h, and they were kept frozen until DNA extraction. The other samples (16 %), including 4 hair samples, were placed in a 50 mL plastic tube, preserved by drying with silica beads (Wasser et al. 1997), and stored at ambient temperature until DNA extraction. DNA extraction from feces, hair, and saliva samples was performed using the GuSCN/silica method (Boom et al. 1990; Höss and Pääbo 1993), ISOHAIR (NIPPON GENE), and the QIAamp DNA mini kit (QIAGEN), respectively (Sugimoto et al. 2012). DNA extraction from all samples, including negative controls, was performed in a dedicated space using dedicated micropipettes with aerosol-resistant tips to avoid contamination.

**Table 1** Summary of samples collected and sampling methods in the present study

| Winter    | No. of sample |               | Sampling period | No. of personnel | Sampling method | Sampling range   |
|-----------|---------------|---------------|-----------------|------------------|-----------------|------------------|
| 2000–2001 | 15            | (13F, 1H, 1S) | Jan–Mar, May    | 4                | OP, FT          | –                |
| 2001–2002 | 102           | (91F, 6H, 5S) | Nov–Mar         | 30               | OP, FT          | LR               |
| 2002–2003 | 104           | (102F, 2H)    | Nov–Feb         | 30               | OP, FT          | LR               |
| 2004–2005 | 65            | (37F, 28H)    | Jan–Apr         | 4                | OP, FT, HT      | HDA              |
| 2005–2006 | 12            | (12F)         | Mar             | 2                | SDD             | –                |
| 2006–2007 | 86            | (80F, 6H)     | Oct, Dec–May    | 10               | OP, FT, SDD     | LR excluding HDA |
| 2007–2008 | 88            | (87F, 1H)     | Dec–Mar         | 16               | OP, FT          | LR               |

*F* feces, *H* hairs, *S* saliva, *OP* opportunistic, *FT* following the tracks of large cats, *HT* hair traps, *SDD* scat detection dogs, *LR* leopard range except the southernmost area, *HDA* high track density area

## DNA analyses

Feces, hair, and saliva samples collected from the field were potentially derived from other sympatric carnivores, and thus, the samples first required species identification. Species and sex identification was performed following the method of Sugimoto et al. (2006), where species was identified by amplifying a part of the cytochrome *b* gene in the mitochondrial DNA (mtDNA), and sex was identified by amplifying a part of the DBY gene on the Y chromosome and/or the ZFX gene on the X chromosome. For individual identification, we selected 13 dinucleotide microsatellite loci (FCA008, FCA026, FCA043, FCA090, FCA096, FCA097, FCA098, FCA105, FCA123, FCA211, FCA224, FCA229 and FCA247) (Menotti-Raymond et al. 1999) of the 25 microsatellite loci reported by Uphyrkina et al. (2002) on the basis of multilocus genotypes obtained by amplifying the DNA derived from two wild caught, eight captive individuals, and seven good quality fecal samples which were presumably from different individuals based on their sampling location. Marker selection was based on the following criteria: (1) amplifying short lengths of DNA, (2) ease of amplification, (3) the presence of many alleles, and (4) ease of scoring. Genotyping of the leopard samples was performed using multiplex PCR, with the following loci combinations: (1) FCA008, FCA026, FCA098; (2) FCA097, FCA211, FCA224; (3) FCA043, FCA105; (4) FCA090, FCA123; (5) FCA229, FCA247; (6) FCA096. The microsatellite loci were amplified in 10  $\mu$ L volumes including 1 $\times$  PCR buffer, 0.2 mM each dNTP, 3.0 mM MgCl<sub>2</sub>, 0.4  $\mu$ M of each primer, 4  $\mu$ g of BSA, 0.3 units of *Taq* DNA polymerase (Applied Biosystems), and 1.0  $\mu$ L of DNA extract. The reaction conditions were those described by Menotti-Raymond et al. (1999), although we increased the number of second cycles from 20 to 30, i.e. 40 cycles in total. The PCR products were analyzed with an ABI 3100 automatic sequencer (Applied Biosystems) and the genotyping data were collected using GeneScan (Applied Biosystems).

We performed genotyping at least five times (three times for the hair samples) for samples collected in the first three winters (2000–2001, 2001–2002, and 2002–2003). A heterozygous genotype was determined when both alleles were detected at least three times (two times for hair), whereas a homozygote was determined when one allele was detected at least five times (three times for hair). Based on these genotyping data, we calculated the genotyping error rates for each locus using equations 1 and 3 from Broquet and Petit (2004). The error rates varied across loci and fell within the range of 0–0.185 for allelic dropout and 0–0.026 for false alleles, whereas the average error rates for allelic dropout and false alleles were 0.084 and 0.010, respectively (Table 2). Based on these error rates, we

considered that three amplifications could effectively minimize genotyping errors. Therefore, for samples from the last four winters (2004–2005, 2005–2006, 2006–2007, and 2007–2008), we performed genotyping at least three times, and heterozygous and homozygous genotypes were determined when they were observed at least two and three times, respectively.

To examine nucleotide diversity of the individuals identified, we sequenced a part of the cytochrome *b* gene (846 bp). Two partially overlapped fragments were amplified separately using four primers, three of which were developed in the present study. The first half (498 bp) was amplified by PSF1 (5'-ACTCATTCATTGATCTCCC CGCT-3') and PSR1 (5'-GGAAGGCAAAGAATCGTGT CAA-3'), and the second half (440 bp) was amplified by PSF2 (5'-GGCTTCTCAGTAGACAAAGCT-3') and PH1 (Nagata et al. 2005). The composition of the PCR mixture was the same as that described in the species identification (Sugimoto et al. 2006) and the reaction conditions were 94 °C for 10 min followed by 40 cycles of 30 s at 94 °C, 45 s at 55 °C, 1 min at 72 °C, and a final extension for 10 min at 72 °C. PCR products were directly sequenced in both forward and reverse directions using a BigDye Terminator Kit ver. 1.1 (Applied Biosystems) and an ABI 3100 automatic sequencer (Applied Biosystems).

**Table 2** Genetic variation at 13 microsatellite loci for the 32 individuals identified during the five recent winters (2002–2003 to 2007–2008)

| Locus   | A    | $A_E$ | $H_E$ | $H_O$ | $P_{ID-sib}/locus$ | ADO   | FA    |
|---------|------|-------|-------|-------|--------------------|-------|-------|
| FCA026  | 4    | 2.36  | 0.58  | 0.63  | 0.52               | 0.020 | 0.005 |
| FCA229  | 3    | 2.26  | 0.57  | 0.56  | 0.54               | 0.078 | 0.025 |
| FCA096  | 3    | 2.25  | 0.56  | 0.47  | 0.55               | 0.147 | 0.016 |
| FCA211  | 3    | 2.17  | 0.55  | 0.47  | 0.56               | 0.057 | 0.003 |
| FCA098  | 2    | 1.99  | 0.51  | 0.56  | 0.60               | 0     | 0.005 |
| FCA043  | 2    | 1.93  | 0.49  | 0.50  | 0.60               | 0.061 | 0     |
| FCA090  | 2    | 1.82  | 0.46  | 0.56  | 0.63               | 0.033 | 0     |
| FCA247  | 2    | 1.60  | 0.38  | 0.31  | 0.68               | 0.185 | 0.010 |
| FCA224  | 2    | 1.60  | 0.38  | 0.44  | 0.68               | 0.146 | 0.009 |
| FCA105  | 2    | 1.48  | 0.33  | 0.34  | 0.72               | 0.083 | 0.003 |
| FCA008  | 3    | 1.37  | 0.27  | 0.31  | 0.76               | 0.040 | 0.002 |
| FCA097  | 3    | 1.29  | 0.23  | 0.22  | 0.79               | 0.081 | 0.026 |
| FCA123  | 3    | 1.29  | 0.23  | 0.25  | 0.79               | 0.072 | 0.017 |
| Average | 2.62 | 1.80  | 0.43  | 0.43  |                    | 0.084 | 0.010 |

The genotyping error rates for ADO and FA were calculated based on the samples in the first three winters (2000–2001 to 2002–2003). Loci are ranked from low to high values of  $P_{ID-sib}/locus$

A observed number of alleles,  $A_E$  effective number of alleles,  $H_E$  expected heterozygosity,  $H_O$  observed heterozygosity,  $P_{ID-sib}/locus$  probability of identity for siblings per locus, ADO allelic dropout, FA false alleles

## Data reliability

To improve the reliability of the genetic data, we considered potential sources of errors and performed additional experiments as required. The presence of DNA of leopard cat (*Felis bengalensis*) in the fecal DNA extracts, which is one of the prey for the leopard (Aramilev et al. unpublished data), could lead to unreliable genotyping because the microsatellite markers used are applicable to a broad range of cat species (Menotti-Raymond et al. 1999). To exclude this possibility, we screened all samples before genotyping was performed. The screening involved PCR using the leopard cat-specific primers developed in the present study (FB-CbF: 5'-CTCACATCTGTCGTGACGTT-3'; FB-CbR: 5'-TGGCTATGACTGCGAATAGC-3'), which amplified a part of the leopard cat cytochrome *b* gene (177 bp). If amplifications were observed, we removed those DNA extracts and re-extracted the fecal DNA using other parts of the fecal sample, assuming that the DNA of the prey species would not be evenly distributed within the feces. Only samples with negative amplifications were used in the subsequent genotyping process. PCR for the detection of leopard cat DNA was performed in a 15  $\mu$ L volume containing 1 $\times$  PCR buffer, 0.2 mM each dNTP, 3.0 mM MgCl<sub>2</sub>, 0.5  $\mu$ M of each primer, 9  $\mu$ g of BSA, 0.4 units of *Taq* DNA polymerase (Applied Biosystems) and 1.0  $\mu$ L of fecal extracts, and the reaction condition was 94 °C for 10 min followed by 35 cycles of 30 s at 94 °C, 30 s at 55 °C, 30 s at 72 °C and a final extension for 5 min at 72 °C. 4  $\mu$ L of the PCR products was electrophoresed on a 2 % agarose gel.

A series of DNA analyses (DNA extraction, species/sex identifications, and genotyping) were performed to verify the genotype in instances where a single sample represented a distinctive genotype. Genotyping was performed at least three times during this second DNA analyses. In addition, because genotyping errors create “false individuals” that differ from others by one or two loci (Paetkau 2003), we performed genotyping two times for the relevant loci of the pairs of genotypes. The resulting individual genotypes were further examined by the program MICROCHECKER ver. 2.2.3 (Van Oosterhout et al. 2004) to avoid scoring errors due to stuttering or large allele dropout.

We calculated the probability of identity for unrelated individuals ( $P_{ID-unrelate}$ ) and siblings ( $P_{ID-sib}$ ) (Waits et al. 2001) using the program GIMLET ver. 1.3.3 (Valière 2002) to assess whether the number of markers was sufficient to identify different individuals. We did not know the a priori allele frequencies in the population, and thus, we calculated the probability of identities from the genotyping data of the noninvasive genetic samples. These values were also calculated for the winters of 2002–2003 and

2007–2008 separately, for which population size estimations were performed (see below). To assess the marker power in the present study, we plotted the genotype similarity distribution, i.e. the number of mismatched loci, in each pair of individuals identified. We also used this distribution to verify whether it had a unimodal or bimodal distribution because the latter indicates the possibility of genotyping errors (McKelvey and Schwartz 2004).

## Population size estimation

Population size was estimated using genotyping data from two winters (2002–2003 and 2007–2008) when we collected a large number of samples and covered the majority of leopard habitat. Because our sampling did not consider capture/recapture sessions, we used the Capwire model, which is applicable to a non-session-based dataset (Miller et al. 2005). Population size was estimated using the two innate rates model in the program CAPWIRE (Miller et al. 2005) because it was unlikely that individuals would have equal capture probabilities, given that the number of fecal samples per individual differed between sexes (see “Results” section). The Capwire model assumes that the population is closed during the sampling period. However, our samples from 2002–2003 to 2007–2008 were collected over about 3 months, and the leopards were capable of traveling long distances, particularly males (Augustine et al. 1996). Therefore, the closed assumption was unlikely to have held strictly in the present study, which may have led to an upward bias when estimating the population size. Leopard tracks were found in areas we did not sample, including the Chinese area near the border (Yang et al. 1998) and the southernmost area of Southwest Primorskii Krai (Pikunov et al. 2003, 2009). In both places, the track density was much lower than that in the area we sampled, although some individuals may have moved in/out of the sampling area during our sampling period. Given existing knowledge of track locations and track density, dispersal of individuals was probably rare, unlikely to have caused a significant overestimation, but our estimates should be treated with caution.

## Genetic diversity and $N_e$ estimation

Genetic diversity was examined based on microsatellite variation in nuclear DNA and nucleotide diversity in mtDNA. Using individual genotypes identified during the five recent winters (2002–2003 to 2007–2008), the number of alleles per locus and the expected and observed heterozygosity were calculated using the program GENEPOP 3.4 (Raymond and Rousset 1995). The effective number of alleles per locus was also calculated according to the formula in Kimura and Crow (1964). Hardy–Weinberg

equilibrium and linkage disequilibrium were tested using the program GENEPOP 3.4, where a sequential Bonferroni correction was used for multiple tests (Rice 1989). We compared the genetic diversity, i.e. the expected heterozygosity ( $H_E$ ) and allelic richness ( $AR$ ), between the individuals identified during the earlier winters (2001–2002 and 2002–2003) and those identified during the later winters (2006–2007 and 2007–2008) to test for any increase/decrease in genetic diversity during the study period. For this comparison, we used a Wilcoxon signed rank test with loci as the pairing factor.  $AR$ , which was corrected with the smallest sample size, was calculated using the program FSTAT ver. 2.9.3.2 (Goudet 2001). We also compared the allelic compositions of the 13 loci of the individuals identified in the present study with those of 2 individuals captured between 1993 and 1996 to assess whether there was any possible loss of allelic diversity.

We estimated effective population sizes ( $N_e$ ) for two time periods (2001–2002 to 2002–2003 and 2006–2007 to 2007–2008) using the individual genotypes identified in each time period. We used two genetic-based methods. The first method estimated  $N_e$  from data on linkage disequilibrium. This method is implemented in the program LDN<sub>e</sub> (Waples and Do 2008). We selected the estimates when excluding all alleles with frequencies of  $<0.05$ . The second method estimated  $N_e$  on the basis of eight summary statistics using an approximate Bayesian computation. This method is implemented in the program ONeSAMP (Tallmon et al. 2008). We set the prior range for  $N_e$  as 2–100. To meet the assumption of no population structure for the  $N_e$  estimations, we conducted a Bayesian clustering analysis (Pritchard et al. 2000) for each separate time period and confirmed that the population in Southwest Primorskii Krai was a single genetic cluster (data not shown).

## Results

### DNA analyses

We collected 472 samples (422 feces, 44 hairs, and 6 saliva) during seven winters (Table 1). We identified 239 samples (224 feces, 12 hairs, and 3 saliva) as being derived from leopards. Of these, 155 samples (151 feces, 4 hairs, and 0 saliva) were successfully genotyped at the 13 loci. The success rate of individual identification varied among winters, from 33.3 % in 2000–2001 to  $>80$  % in 2002–2003/2007–2008, with an average of 64.9 % (155/239). We did not observe any Bengal leopard cat-specific amplifications among these genotyped samples. Comparison of the 155 multilocus genotypes allowed us to identify 38 individuals (18 males and 20 females). However, one female identified from a single hair sample collected in 2000–2001 was not reconfirmed in

the second DNA analyses because most of the sample was used in the first DNA extraction and few hair strands remained. Therefore, we excluded this female individual from our dataset, and subsequent analyses were performed using 37 individuals (18 males and 19 females). The individuals were identified from 1 to 39 different times (Table 3). Twenty-two individuals (10 males and 12 females) were identified in a single winter, whereas 5 individuals (4 males and 1 female) were identified over 5 or 6 years (Table 3). We found that the number of fecal samples per male was significantly higher than that per female individual (mean  $\pm$  SD =  $6.7 \pm 9.6$  for males,  $1.6 \pm 0.8$  for females; Brunner–Munzel test:  $P = 0.046$ ). Almost half of the individuals were identified from a single sample each, but, two male individuals, i.e. ML3 and ML9, were identified from large numbers of samples (Table 3), and we illustrated the locations of these multiple samples from 2007 to 2008 to identify their spatial distributions (Fig. 2). Interestingly, their distributions overlapped extensively. Another male, ML6, also appeared to use the same habitat, although only two locations could be derived from him. In 2002–2003, the distribution of male individuals was less clear because of the small number of locations, but two males, i.e. ML5 and ML6, appeared to have partially overlapping home ranges in the same area.

The genotype similarity distribution was unimodal with a peak at 8 mismatched pairs, and there were zero 1 mismatched and two 2 mismatched pairs (Fig. 3). Sex was different in one of the two 2 mismatched pairs. In the other pair, one allele differed at a locus where both individuals had a heterozygous genotype, and each of the two individuals was identified from 10 to 6 samples, respectively. Therefore, the two 2 mismatched pairs were unlikely to be related to genotyping errors. Among the multiple samples from each individual, we found no inconsistent sex identifications, i.e. 39 samples from ML3 were all assigned as males, whereas three samples from FL1 were all assigned as females. The resultant  $P_{ID-unrelate}$  and  $P_{ID-sib}$  were  $4.02 \times 10^{-6}$  and  $2.87 \times 10^{-3}$ , respectively ( $P_{ID-unrelate} = 2.6 \times 10^{-6}$ ,  $P_{ID-sib} = 3.3 \times 10^{-3}$  in 2002–2003;  $P_{ID-unrelate} = 2.4 \times 10^{-6}$ ,  $P_{ID-sib} = 3.2 \times 10^{-3}$  in 2007–2008).

### Population size estimation

The population size was estimated as 28 (95 % CI 19–38) in 2002–2003 and 26 (95 % CI 13–33) in 2007–2008, indicating a slightly lower estimate for 2007–2008, although the 95 % CI overlapped to a large extent.

### Genetic diversity and $N_e$ estimation

The microsatellite variation of the 32 individuals identified during the 5 recent winters showed that the number of

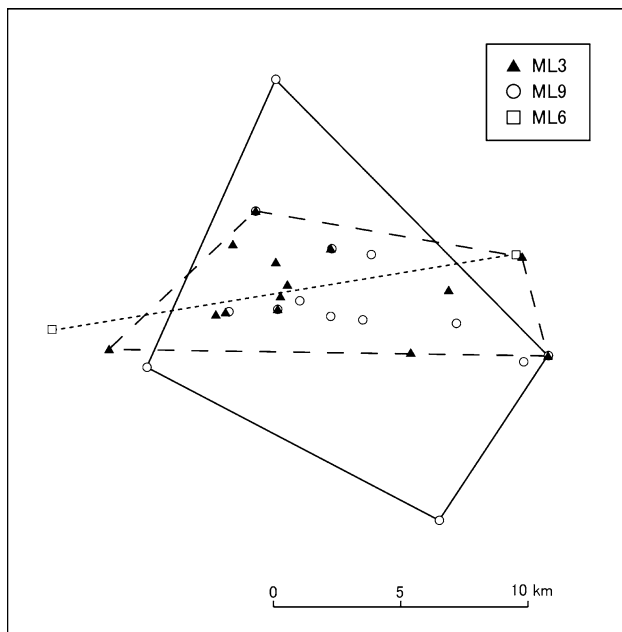
**Table 3** Results of individual identification for the noninvasive genetic samples collected during the seven winters

| Individuals       | Total no. of observation | Winters observed     |                       |                       |                       |                      |                       |                       |
|-------------------|--------------------------|----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|-----------------------|
|                   |                          | 2000–2001<br>(n = 9) | 2001–2002<br>(n = 59) | 2002–2003<br>(n = 58) | 2004–2005<br>(n = 31) | 2005–2006<br>(n = 9) | 2006–2007<br>(n = 24) | 2007–2008<br>(n = 49) |
| ML1               | 1                        | 1                    |                       |                       |                       |                      |                       |                       |
| ML2               | 4                        |                      | 4                     |                       |                       |                      |                       |                       |
| ML3               | 39                       |                      | 10                    | 8                     | 5                     | 2                    |                       | 14                    |
| ML4               | 4                        |                      | 2                     | 2                     |                       |                      |                       |                       |
| ML5               | 13                       |                      | 4                     | 9                     |                       |                      |                       |                       |
| ML6               | 11                       |                      | 2                     | 4                     | 3                     |                      |                       | 2                     |
| ML7               | 10                       |                      |                       | 2                     |                       |                      | 7                     | 1                     |
| ML8               | 6                        |                      |                       | 6                     |                       |                      |                       |                       |
| ML9               | 19                       |                      |                       | 1                     |                       | 4                    |                       | 14                    |
| ML10              | 2                        |                      |                       | 2                     |                       |                      |                       |                       |
| ML11              | 1                        |                      |                       | 1                     |                       |                      |                       |                       |
| ML12              | 5                        |                      |                       | 1                     |                       |                      | 4                     |                       |
| ML13              | 1                        |                      |                       | 1                     |                       |                      |                       |                       |
| ML14              | 1                        |                      |                       | 1                     |                       |                      |                       |                       |
| ML15              | 1                        |                      |                       |                       | 1                     |                      |                       |                       |
| ML16              | 2                        |                      |                       |                       |                       |                      | 1                     | 1                     |
| ML17              | 1                        |                      |                       |                       |                       |                      | 1                     |                       |
| ML18              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| FL1               | 3                        | 1                    | 2                     |                       |                       |                      |                       |                       |
| FL2               | 1                        |                      | 1                     |                       |                       |                      |                       |                       |
| FL3               | 3                        |                      | 1                     | 1                     | 1                     |                      |                       |                       |
| FL4               | 2                        |                      | 1                     | 1                     |                       |                      |                       |                       |
| FL5               | 1                        |                      | 1                     |                       |                       |                      |                       |                       |
| FL6               | 2                        |                      |                       | 2                     |                       |                      |                       |                       |
| FL7               | 3                        |                      |                       | 2                     | 1                     |                      |                       |                       |
| FL8               | 2                        |                      |                       | 1                     |                       |                      |                       | 1                     |
| FL9               | 3                        |                      |                       | 2                     | 1                     |                      |                       |                       |
| FL10              | 1                        |                      |                       | 1                     |                       |                      |                       |                       |
| FL11              | 3                        |                      |                       |                       | 2                     |                      |                       | 1                     |
| FL12              | 1                        |                      |                       |                       |                       |                      | 1                     |                       |
| FL13              | 1                        |                      |                       |                       |                       |                      | 1                     |                       |
| FL14              | 1                        |                      |                       |                       |                       |                      | 1                     |                       |
| FL15              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| FL16              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| FL17              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| FL18              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| FL19              | 1                        |                      |                       |                       |                       |                      |                       | 1                     |
| Total             | 154                      | 2                    | 28                    | 48                    | 14                    | 6                    | 16                    | 40                    |
| (n feces/n hairs) | (151/3)                  | (2/0)                | (27/1)                | (48/0)                | (13/1)                | (6/0)                | (16/0)                | (40/0)                |

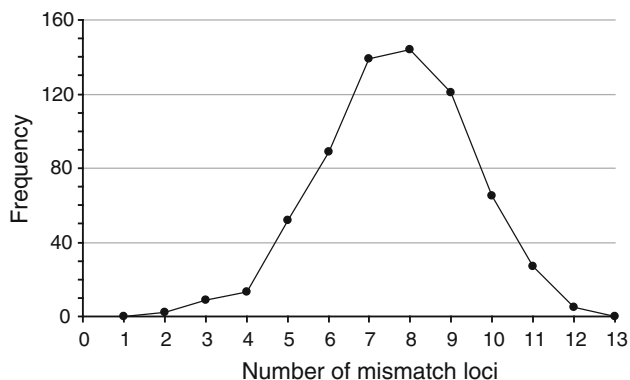
The individuals were labeled as ML# (male leopard no.) and FL# (female leopard no.) in the order of their sampling date. Numbers in parentheses below the winters observed indicate the number of leopard samples revealed from species identification

alleles per locus ranged from 2 to 4 with an average of 2.62, and the average expected and observed heterozygosity were both 0.43 (Table 2). The effective number of alleles per locus was lower than the observed number of

alleles, ranging from 1.29 to 2.36 with an average of 1.80 (Table 2). We found no significant deviation from Hardy–Weinberg or linkage equilibrium at the 13 loci. Comparison of the genetic diversity ( $H_E$  and  $AR$ ) for the 23



**Fig. 2** Home range overlap of two male individuals, ML3 and ML9, observed in the high track density area in 2007–2008. Locations of ML6 were also shown. Home ranges of ML3 (dash line) and ML9 (solid line) were obtained by a minimum convex polygon, and a movement path of ML6 were drawn by a dotted line



**Fig. 3** Mismatch distribution of the genotype similarity between all pairs of the 37 individuals identified

individuals identified during the earlier winters (2001–2002/2002–2003) and the 18 individuals during the later winters (2006–2007/2007–2008) showed there was only a slight decrease with no significant difference ( $H_E = 0.43$ ,  $AR = 2.57$  for earlier winters;  $H_E = 0.42$ ,  $AR = 2.46$  for later winters; Wilcoxon signed rank test:  $P = 0.787$  for  $H_E$ ,  $P = 1.00$  for  $AR$ ). We found that one allele at the FCA105 locus, which was present in a leopard captured in the mid-1990s, was not present in the 37 individuals identified in the present study. We may have missed sampling this allele, but its absence from 37 individual genotypes may indicate a loss of allelic diversity.

Sequencing of a part of the cytochrome *b* gene (846 bp) showed two haplotypes, which differed by an A/T transversion at position 490. However, four individuals had both A and T peaks, suggesting the presence of heteroplasmy (A:  $n = 8$ , T:  $n = 25$ , both A and T:  $n = 4$ ). We recognized this as heteroplasmy rather than a nuclear copy of the mitochondrial sequence because a double peak was observed only at this site, and the sequences were comparable to published mitochondrial cytochrome *b* sequences of the Far Eastern leopard (AB211401–AB211407). Sequences of the two haplotypes were deposited in GenBank (AB817078, AB817079).

For the  $N_e$  estimations, we used 23 individual genotypes in the earlier time period (2001–2002 to 2002–2003) and 18 individual genotypes in the later time period (2006–2007 to 2007–2008). The linkage disequilibrium method ( $LDN_e$ ) estimated  $N_e$  in the earlier and later time periods as 12.3 (95 % CI 5.7–30.9) and 7.8 (95 % CI 2.7–23.4), respectively, whereas the approximate Bayesian computation method (ONeSAMP) estimated them as 16.1 (95 % credible limits 13.8–19.6) and 14.0 (95 % credible limits 11.9–17.6), respectively.

## Discussion

### DNA analyses

The noninvasive genetic samples collected during winter in this region provided amplifiable DNA that was sufficient to identify species, sex, and individuals. The higher success rates of individual identification in 2002–2003/2007–2008 were probably due to the shorter time interval between sample collection and DNA extraction and better sample handling to prevent the frozen samples from thawing. During the first 2 winters, the samples were retained for over 3 years before DNA extraction and some frozen samples were not adequately treated in the field or during transport (Sugimoto et al. 2012). If we exclude the samples from the first 2 winters, the success rate was 72.5 %. Therefore, adequate sample handling and immediate DNA extraction after sample collection are important for increasing the success rate of nuclear DNA amplification. Of the 26 leopard fecal samples preserved by the drying method with silica beads, 17 were successfully used for individual identification (67.2 %). There was no significant difference in the success rate of individual identification between the two methods of feces preservation (freezing method and drying method with silica beads; Fisher's exact test:  $P = 0.219$ ), but we recommend using the freezing method because the success rate was >80 % when the sample was handled properly. The drying method with

silica beads may be more useful in situations where freezing is not an option.

Eighteen of 37 individuals were each identified from a single sample. This small number of samples per individual may be due to several factors. First, we used an opportunistic sampling approach and sampling was not systematically repeated, and some areas were better surveyed than others. This unequal sampling intensity probably affected the detection frequency of individuals. Second, individual identification was not always successful. 35 % of leopard samples (84/239) were not genotyped successfully at the 13 loci probably because they were old or degraded. Third, there was a sex-bias when collecting samples. Of the 18 individuals with a single sample, 11 were females.

The reason for the higher number of male fecal samples may be related to greater mobility and larger home range of males (Augustine et al. 1996) or a greater propensity for males to leave scent marks in prominent locations (Salmanova 2011). Given this difference between sexes, it is probable that our sampling missed more females, and thus, the remaining population may have a female-biased sex ratio, although our individual identification results indicated that the population was male-biased in 2002–2003 (1.71 M:1F) and had an almost equal sex ratio in 2007–2008 (1 M:1.17F).

Spatial distribution of scats from several individuals suggests that it may be common for home ranges of males to overlap in the study area. Previous studies of leopard spatial ecology in Asia and Africa have suggested territoriality (mutually exclusive or almost exclusive home ranges) in some areas (e.g. Rabinowitz 1989; Mizutani and Jewell 1998; Odden and Wegge 2005), while in other areas male leopards had widely overlapping home ranges (e.g. Jenny 1996; Stander et al. 1997; Marker and Dickman 2005). These differences are probably due to variable environmental conditions, such as the habitat type or prey availability, as well as methodological differences. Marker and Dickman (2005) noted that variations in spatial utilization among studies reflected the leopard's ecological flexibility, which may allow the leopard to adapt to various environments throughout its vast range from Africa to Asia. In the area where we observed home range overlaps, i.e. the high track density area, the original mixed forest remained, which mainly comprised Korean pine, black fir, and Mongolian oak, and the prey density was high, thereby indicating that this area is one of the most suitable habitats for the leopard (Miquelle and Murzin 2000). Therefore, high prey abundance in the area compared with that in surrounding areas may be a possible cause for the home range overlaps. This may indicate that, if an area has high habitat quality, it may be able to accommodate more males than expected from the home range size with their spatial exclusivity. The limited number of locations we recorded

during the winter, however, failed to adequately quantify the degree of home range overlap and its seasonality, although our general conclusions about the overlapping home ranges of males have also been suggested for radio-collared individuals as well (Salmanova 2008).

#### Population size estimation

Our estimates did not correspond to the entire population size because we did not conduct sampling in the southernmost part of the leopard habitat. Given the low track density in that area, it is likely that only a small number of individuals were present, and thus, the population size estimate for the entire leopard distribution could be only slightly higher than our estimates. Snow-track counts conducted in February 2003 (winter 2002–2003) and February 2007 (winter 2006–2007) were based on approximately 150 established transects that covered the entire range of leopards in Southwest Primorskii Krai, resulting in population size estimates of 28–30 in 2003 and 25–34 in 2007 (Pikunov et al. 2003, 2009). Although our estimates had wider intervals, the point estimates were quite similar to those of the snow-track counts, and both methods produced similar population size estimates in two different winters. Our estimate for 2002–2003 was also comparable to the estimate from the camera trap capture–recapture study in the same winter, 33 (95 % CI 29–38), which extrapolated the density estimates from a study area over the entire range of leopards based on the snow-track data of Pikunov et al. (2003) (Kostyria et al. 2003).

The present study produced similar population size estimates for 2002–2003 and 2007–2008, although our comparison should be treated with care because the sampling intensity was not the same between the two winters. Because we used opportunistic sampling, it is difficult to accurately estimate sampling intensity in each of two winters. However, given the number of people involved, the sampling intensity was higher in 2002–2003, particularly in areas with lower track densities. This may have led to a larger number of individual detections and a slightly larger population size estimate in 2002–2003. In addition, the capture heterogeneity in 2007–2008 was exceptionally high compared with that in 2002–2003 (Table 3). Although the Capwire model works well in the presence of capture heterogeneity, extremely high capture heterogeneity leads to underestimation (Miller et al. 2005). For this reason, the population size estimate for 2007–2008 is probably less reliable than that for 2002–2003. The main cause of the high capture heterogeneity in 2007–2008 was probably the unequal sampling intensity. An equal sampling intensity across the entire leopard habitat would likely increase the number of individuals detected and the average number of observations per individual without extreme capture

heterogeneity, thereby leading to a more reliable estimate with a narrower CI (Miller et al. 2005).

This is the first case in which the leopard population size was estimated by two independent approaches (a DNA-based approach and a track-based approach). Similar results obtained from both approaches may provide more confidence that the leopard population size changed little during the study period.

#### Genetic diversity and $N_e$ estimation

Given the number of alleles, heterozygosity, and the number of mtDNA haplotypes, it is clear that the remaining leopard population has low genetic diversity. Similar results were reported by Uphyrkina et al. (2002), who examined 25 microsatellite loci (including the 13 loci we used) and part of the mtDNA (611 bp of NADH-5 and 116 bp of Dloop) for 7 individuals captured in Southwest Primorskii Krai between 1993 and 1996. They showed that the average number of alleles per locus, the average effective number of alleles per locus, and the average observed heterozygosity were 2.32, 1.7, and 0.402, respectively, and they detected one mtDNA haplotype. In the present study, the amount of genetic diversity was slightly higher than that reported by Uphyrkina et al. (2002), but given that variable loci were selected and monomorphic loci were not included in the present study, both studies appear to have essentially produced the same results.

The effective population size estimated by both methods was very low, ranging from 7 to 16 in the point estimates, but this is not surprising given the small population size estimates. Both estimation methods yielded slightly lower estimates for the later time period, although there was substantial overlap of the 95 % CIs between the two periods. When we used the point estimates of the two  $N_e$  estimation methods for each time period and the upper and lower bounds of the population size estimates from the snow-track counts in the corresponding winters, i.e. 2002–2003 (28–30) and 2006–2007 (25–34), the  $N_e/N$  ratios in the earlier and later time periods were 0.41–0.44 and 0.23–0.31 with  $LDN_e$ , respectively, and 0.54–0.58 and 0.41–0.56 with  $ONeSAMP$ , respectively. These ratios were all higher than the mean empirical  $N_e/N$  ratio, 0.11 (Frankham 1995), but they agreed with the finding that the  $N_e/N$  ratio was higher with a lower population size, presumably due to a genetic compensation effect (e.g. Ardren and Kapuscinski 2003). A higher  $N_e$  relative to  $N$  in a small population due to intrapopulation biological mechanisms may be beneficial in terms of reducing the loss of genetic variation. However, a high  $N_e/N$  ratio itself would be of limited significance in improving the critical situation for the leopard population.

A lack of information about leopards in nearby China or North Korea until the late 1990s makes it difficult to determine whether the population in Southwest Primorskii Krai could have been affected by natural immigration from adjacent areas. However, given that the population size has remained at a low level (22–50) since 1972, genetic drift and inbreeding would have had a certain effect on this population in the past and a stronger effect in recent years. Although we did not find a significant decline in  $H_E$  and  $AR$  during the study period, we found an indication of the loss of one allele at the FCA105 locus. Alleles with a low frequency will be lost, whereas common alleles will be fixed. The state of alleles with a moderate frequency will also change considerably in a small population because their fates are strongly affected by stochastic properties, which makes it possible that deleterious alleles could become fixed in the population (drift load). In addition, the loss of heterozygosity per generation will inevitably occur. If the population remains constant at  $N_e$  of 14 (harmonic mean of the  $ONeSAMP$  point estimates) during the period between 2001–2002 and 2007–2008 (i.e. equivalent to 1–2 generations given the age of first reproduction is about 3 years, Sunquist and Sunquist 2002), the rate of loss of heterozygosity per generation would be 3.6 % (based on  $H_t = H_0(1 - 1/2N_e)^t$ , Crow and Kimura 1970). This effect could be associated with the slightly lower, although not significant, value of  $H_E$  observed during the later winters. Nonetheless, a recent medical survey of four captured individuals found no apparent evidence of inbreeding depression (Lewis 2010). It should be noted, however, that these results were not conclusive because the sample size was small and lethal and sublethal effects are often expressed early in development, which means that inbreeding depression is not always detectable (Keller and Waller 2002).

#### Conservation implications

The present study demonstrated that noninvasive genetic sampling represents an effective means for obtaining ecological and genetic data for elusive and endangered species such as leopards and that this approach represents a viable population monitoring tool even for species with low genetic diversity. Our results indicate that the genetic variations in polymorphic loci were limited, effective population size was small, and there were no signs of improvement in genetic diversity, which suggests that the remaining population will continue to suffer loss of genetic diversity. If evidence of inbreeding depression is observed, introduction of captive individuals, who have a higher genetic diversity than the wild population (Uphyrkina et al. 2002), needs to be considered. In addition, because leopard tracks generally occur throughout all suitable habitats (Miquelle and Murzin 2000) and population size estimates

have been stable since the 1970s, expanding the available and suitable habitat is essential for increasing population size, thereby mitigating genetic erosion and improving the tolerance of demographic, environmental, and artificial stochastic events. Potentially suitable habitat for population expansion exists in China across the border. Recent photographic evidence of a female leopard with two cubs some 30 km from the border provides great hope that protective efforts in the Hunchun and neighboring Wangqing areas of Jilin Province are bringing results. In addition to conserving and expanding the remaining population, the creation of a second population in a former habitat using captive individuals may be critical for ensuring the persistence of the Far Eastern leopard in the wild (Hebblewhite et al. 2011). If a second population can be successfully established, artificial migration (i.e. translocation) may be possible from the second to the original population, which would improve the genetic and demographic status of the recipient population while minimizing the possibility of outbreeding depression because any translocated individuals will already have survived in a natural environment.

**Acknowledgments** We thank the Kobe Oji Zoo, Hiroshima City Asa Zoological Park, Asahiyama Zoo, and S. J. O'Brien (the National Cancer Institute, USA) for the Far Eastern leopard DNA samples, and Inokashira Zoo, R. Masuda, and Ueno Zoological Gardens for the leopard cat DNA samples. We also thank M. Stuewe, Y. Uryu, K. Kobayakov, Y. Darman, O. Walter, and S. Christie for providing us with valuable information on the Far Eastern leopard; and the WWF Russian Far East office for providing us with maps and information on the forest restoration project. We express special gratitude to A. Belozor and S. Higashi for their assistance in the realization of this project. Financial support for this study was provided by the A. Starker Leopold Endowed Chair, University of California, Berkeley (D. McCullough, chair-holder) and a Grant-in-Aid for JSPS Fellows (19-05514) from Japan Society for the Promotion of Science and in part by WWF US.

## References

Abramov VK, Pikunov DG (1974) Leopards in the far east of the USSR and their protection. Bull Mosc Soc Nat Amateur Sec Biol 79:5–15 in Russian

Adams JR, Vucetich LM, Hedrick PW, Peterson RO, Vucetich JA (2011) Genomic sweep and potential genetic rescue during limiting environmental conditions in an isolated wolf population. Proc R Soc B-Biol Sci 278:3336–3344

Aramilev VV, Fomenko PV (2000) Distribution and number of Far Eastern Leopards in the Southwest Primorski Krai. In: Vas-hukevich YE (ed) Conservation and rational use of animals and floral resources. Irkutsk Agriculture Academy Publishing House, Irkutsk, pp 50–63

Arden WR, Kapuscinski AR (2003) Demographic and genetic estimates of effective population size ( $N_e$ ) reveals genetic compensation in steelhead trout. Mol Ecol 12:35–49

Augustine J, Miquelle DG, Korkishko VG (1996) Preliminary results of the Far Eastern leopard ecology project: implications for conservation and management. Zov Taigi 4:6–11

Boom R, Sol CJA, Salimans MMM, Jansen CL, Wertheimvanden PME, Vandernoordaa J (1990) Rapid and simple method for purification of nucleic acids. J Clin Microbiol 28:495–503

Broquet T, Petit E (2004) Quantifying genotyping errors in noninvasive population genetics. Mol Ecol 13:3601–3608

Crow JF, Kimura M (1970) An introduction to population genetics theory. Harper and Row, New York

De Barba M, Waits LP, Garton EO, Genovesi P, Randi E, Mustoni A, Groff C (2010) The power of genetic monitoring for studying demography, ecology and genetics of a reintroduced brown bear population. Mol Ecol 19:3938–3951

Frankham R (1995) Effective population size/adult population size ratios in wildlife: a review. Genet Res 66:95–107

Goudet J (2001) FSTAT, a program to estimate and test gene diversities and fixation indices

Hebblewhite M, Miquelle DG, Murzin AA, Aramilev VV, Pikunov DG (2011) Predicting potential habitat and population size for reintroduction of the Far Eastern leopards in the Russian Far East. Biol Conserv 144:2403–2413

Höss M, Pääbo S (1993) DNA extraction from Pleistocene bones by a silica-based purification method. Nucl Acids Res 21:3913–3914

Jenny D (1996) Spatial organization of leopards *Panthera pardus* in Tai National Park, Ivory Coast: is rain forest habitat a 'tropical haven'? J Zool 240:427–440

Keller LF, Waller DM (2002) Inbreeding effects in wild populations. Trends Ecol Evol 17:230–241

Kerley LL (2010) Using dogs for tiger conservation and research. Integr Zool 5:390–396

Kimura M, Crow JF (1964) The number of alleles that can be maintained in a finite population. Genetics 49:725–738

Koelewijn HP, Perez-Haro M, Jansman HAH, Boerwinkel MC, Bovenschen J, Lammertsma DR, Niewold FJJ, Kuiters AT (2010) The reintroduction of the Eurasian otter (*Lutra lutra*) into the Netherlands: hidden life revealed by noninvasive genetic monitoring. Conserv Genet 11:601–614

Kostyria AV, Skorodelov AS, Miquelle DG, Aramilev VV, McCullough DR (2003) Results of camera trap survey of Far Eastern leopard population in southwest Primorski Krai, winter 2002–2003. Wildlife Conservation Society, Institute of Sustainable Use of Nature Resources, University of California (Berkeley), Vladivostok, Russia

Lewis J (2010) Disease management program. In: A program for reintroduction of the Far Eastern leopard into southern Sikhote-Alin, Primorskii Krai, Russian Far East. <http://www.wcsrussia.org/en-us/publications/thefareasternleopardscientificpublications.aspx>. Accessed 1 March 2010

MacKay P, Smith DA, Long RA, Parker M (2008) Scat detection dogs. In: Long RA, MacKay P, Zielinski WJ, Ray JC (eds) Noninvasive survey methods for carnivores. Island Press, Washington, pp 183–222

Marker LL, Dickman AJ (2005) Factors affecting leopard (*Panthera pardus*) spatial ecology, with particular reference to Namibian farmlands. S Afr J Wildl Res 35:105–115

McKelvey KS, Schwartz MK (2004) Genetic errors associated with population estimation using non-invasive molecular tagging: problems and new solutions. J Wildl Manage 68:439–448

Menotti-Raymond M, David VA, Lyons LA, Schaffer AA, Tomlin JF, Hutton MK, O'Brien SJ (1999) A genetic linkage map of microsatellites in the domestic cat (*Felis catus*). Genomics 57:9–23

Miller CR, Joyce P, Waits LP (2005) A new method for estimating the size of small populations from genetic mark-recapture data. Mol Ecol 14:1991–2005

Miquelle DG, Murzin AA (2000) Spatial distribution of Far Eastern leopard in Southwest Primorski Krai, and recommendations for

- their conservation. Wildlife Conservation Society, World Wildlife Fund, and Tigris Foundation, Vladivostok
- Miquelle DG, Pikunov DG (2003) Status of the Amur tiger and Far Eastern leopard in Northeast Asia. In: Newell JP (ed) The Russian Far East: a reference guide for conservation and development. Daniel & Daniel Publishers Inc., McKinleyville, pp 106–109
- Miquelle DG, Pikunov DG, Dunishenko YM, Aramilev VV, Nikolaev IG, Abramov VK, Smirnov EN, Salkina GP, Matyushkin EN, Murzin AA (2006) A theoretical basis for surveys of Amur tigers and their prey base in the Russian Far East. DalNauka, Vladivostok in Russian
- Mizutani F, Jewell PA (1998) Home-range and movements of leopards (*Panthera pardus*) on a livestock ranch in Kenya. J Zool 244:269–286
- Nagata J, Aramilev VV, Belozor A, Sugimoto T, McCullough DR (2005) Fecal genetic analysis using PCR-RFLP of cytochrome *b* to identify sympatric carnivores, the tiger *Panthera tigris* and the leopard *Panthera pardus*, in Far Eastern Russia. Conserv Genet 6:863–866
- Nowell K, Jackson P (1996) Wild cats: status survey and conservation management plan. IUCN/SSC Cat Specialist Group, Gland
- Odden M, Wegge P (2005) Spacing and activity patterns of leopards *Panthera pardus* in the Royal Bardia National Park, Nepal. Wildl Biol 11:145–152
- Paetkau D (2003) An empirical exploration of data quality in DNA-based population inventories. Mol Ecol 12:1375–1387
- Pikunov DG, Korkishko VG (1985) Modern distribution and numbers of the leopard (*Panthera pardus*) in the Soviet Far East. Zool J 64:897–905 (in Russian)
- Pikunov DG, Miquelle DG, Abramov VK, Nikolaev IG, Seredkin IV, Murzin AA, Korkishko VG (2003) A survey of Far Eastern leopard and Amur tiger populations in Southwest Primorski Krai, Russian Far East (February 2003). Pacific Institute of Geography (FEB RAS), Wildlife Conservation Society, and Tigris Foundation, Vladivostok
- Pikunov DG, Seredkin IV, Aramilev VV, Nikolaev IG, Murzin AA (2009) Numbers of Far Eastern leopards (*Panthera pardus orientalis*) and Amur Tigers (*Panthera tigris altaica*) in Southwest Primorski Krai, Russian Far East, 2007. DalNauka, Vladivostok
- Pocock RI (1930) The panthers and ounces of Asia. J Bombay Nat Hist Soc 34:307–336
- Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959
- Rabinowitz A (1989) The density and behaviour of large cats in a dry tropical forest mosaic in Huai Kha Khaeng wildlife sanctuary, Thailand. Nat Hist Bull Siam Soc 37:235–251
- Raymond M, Rousset F (1995) Genepop (Version-1.2)—population-genetics software for exact tests and ecumenicism. J Hered 86:248–249
- Rice WR (1989) Analyzing tables of statistical tests. Evolution 43:223–225
- Salmanova E (2008) Territorial use of Far Eastern leopards. M.S. thesis, Far Eastern State University, Primorskii Krai, Russia
- Salmanova E (2011) Characteristics of scrapes of Far Eastern leopards as a primary means of scentmarking. In: Materials from the X conference of students from universities and research organizations of the Russian Far East: problems in ecology, marine biology, and biotechnology, pp 244–245
- Schwartz MK, Luikart G, Waples RS (2007) Genetic monitoring as a promising tool for conservation and management. Trends Ecol Evol 22:25–33
- Stander PE, Haden PJ, Kagece, Ghau (1997) The ecology of a sociality in Namibian leopards. J Zool 242:343–364
- Sugimoto T, Nagata J, Aramilev VV, Belozor A, Higashi S, McCullough DR (2006) Species and sex identification from faecal samples of sympatric carnivores, Amur leopard and Siberian tiger, in the Russian Far East. Conserv Genet 7:799–802
- Sugimoto T, Nagata J, Aramilev VV, McCullough DR (2012) Population size estimation of Amur tigers in Russian Far East using noninvasive genetic samples. J Mamm 93:93–101
- Sun B, Miquelle DG, Xiaochen Y, Zhang E, Hiyai S, Goshen G, Pikunov DG, Dunishenko YM, Nikolaev IG, Daming L (1999) Survey of Amur tigers and Far Eastern leopards in eastern Heilongjiang Province, China, and recommendations for their conservation. Final report to the Wildlife Conservation Society, Vladivostok, Russia
- Sunquist M, Sunquist F (2002) Wild cats of the world. The University of Chicago Press, Chicago
- Taberlet P, Waits LP, Luikart G (1999) Noninvasive genetic sampling: look before you leap. Trends Ecol Evol 14:323–327
- Tallmon DA, Koyuk A, Luikart G, Beaumont MA (2008) ONEsAMP: a program to estimate effective population size using approximate Bayesian computation. Mol Ecol Resour 8:299–301
- Uphyrkina O, Miquelle D, Quigley H, Driscoll C, O'Brien SJ (2002) Conservation genetics of the Far Eastern leopard (*Panthera pardus orientalis*). J Hered 93:303–311
- Valière N (2002) GIMLET: a computer program for analysing genetic individual identification data. Mol Ecol Notes 2:377–379
- Van Oosterhout C, Hutchinson WF, Wills DPM, Shipley P (2004) MICRO-CHECKER: software for identifying and correcting genotyping errors in microsatellite data. Mol Ecol Notes 4:535–538
- Waits LP, Paetkau D (2005) Noninvasive genetic sampling tools for wildlife biologists: a review of applications and recommendations for accurate data collection. J Wildl Manage 69:1419–1433
- Waits LP, Luikart G, Taberlet P (2001) Estimating the probability of identity among genotypes in natural populations: cautions and guidelines. Mol Ecol 10:249–256
- Waples RS, Do C (2008) LDNE: a program for estimating effective population size from data on linkage disequilibrium. Mol Ecol Resour 8:753–756
- Wasser SK, Houston CS, Koehler GM, Cadd GG, Fain SR (1997) Techniques for application of faecal DNA methods to field studies of Ursids. Mol Ecol 6:1091–1097
- Wildlife Conservation Society (2012) Rare appearance for an extremely rare Cat. <http://www.wcs.org/news-and-features-main/rare-appearance-for-extremely-rare-cat.aspx>. Accessed 25 April 2012
- Yang S, Jiang J, Wu Z, Li T, Yang X, Han X, Miquelle DG, Pikunov DG, Dunishenko YM, Nikolaev IG (1998) A survey of tigers and leopards in eastern Jilin Province, China, winter 1998. Final report to the UNDP and the Wildlife Conservation Society, Vladivostok, Russia